

Bond-disordered non-collinear magnets: Perturbative renormalization

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We consider the appropriate continuum modelling for bond disordered non-collinear antiferromagnets and investigate the role of disorder in the fate of long range order in these systems.

I. NL σ -MODEL FOR NON-COLLINEAR ANTIFERROMAGNETS

In non-collinear magnets the original symmetry of the spins is fully broken, e.g. in the triangular lattice heisenberg antiferromagnet the broken symmetry is

$$O(3)/Z_2 \simeq SO(3) \quad (1)$$

.As shown by Azaria et. al [5], for a complete description of the low energy excitation one has to also consider the planar nature of the ground state and distinguish between the three Goldstone modes of the problem. In the following we follow a microscopic analysis [6] to obtain the coarse-grained effective hamiltonian of the triangular lattice antiferromagnet with continuous symmetry. Later, we shall extend this scheme to consdier the effect of bond disorder..

We consider a non-collinear ordered state. The Goldstone modes shall be excited by the generators of the broken symmetry group, G/H . With the trinagular lattice in mind, our continuum modelling involves considering plaquettes of elementary upward triangles, finding the low energy contributions for departure from the 120° order and ultimately coarse-graining within each elementary triangle. We start by denoting the three sublattice orientaions of the 120° order, $\mathbf{n}_A, \mathbf{n}_B$ and $\mathbf{n}_C = -(\mathbf{n}_A + \mathbf{n}_B)$ with spin-excitations generated by $\mathbf{S}_\alpha = U\mathbf{n}_\alpha$, where $U \in G/H$. We consider the hamiltonian within an elementary triangular plaquette.

$$H = J (\mathbf{S}_A(\mathbf{x}_A) \cdot \mathbf{S}_B(\mathbf{x}_B) + \mathbf{S}_B(\mathbf{x}_B) \cdot \mathbf{S}_C(\mathbf{x}_C) + \mathbf{S}_C(\mathbf{x}_C) \cdot \mathbf{S}_A(\mathbf{x}_A)) \quad (2)$$

By performing a gradient expansion coupled with a coarse-graining of the spatial co-ordinate to capture the low energy pieces, we arrive at the following expression.

$$\begin{aligned} H &= JS^2 a^2 (\mathbf{n}_A U^{-1} \cdot \nabla^2 U \mathbf{n}_B - \mathbf{n}_B U^{-1} \cdot \nabla^2 U (\mathbf{n}_A + \mathbf{n}_B) - \mathbf{n}_A U^{-1} \cdot \nabla^2 U (\mathbf{n}_A + \mathbf{n}_B)) \\ &= JS^2 a^2 (\mathbf{n}_A U^{-1} \cdot \nabla^2 U \mathbf{n}_B - (\mathbf{n}_A + \mathbf{n}_B) U^{-1} \cdot \nabla^2 U (\mathbf{n}_A + \mathbf{n}_B)) \end{aligned} \quad (3)$$

The single gradient term cancels from the expansion. Now, a transformation to the orthonormal basis, $\mathbf{e}_1 = (\mathbf{n}_A + \mathbf{n}_B)$ and $\mathbf{e}_2 = \frac{1}{\sqrt{3}}(\mathbf{n}_A - \mathbf{n}_B)$ re-arranges the terms of the hamiltonian in a convenient form.

$$H = -\frac{3JS^2a^2}{2} (\mathbf{e}_1 U^{-1} \cdot \nabla^2 U \mathbf{e}_1 + \mathbf{e}_2 U^{-1} \cdot \nabla^2 U \mathbf{e}_2) \quad (4)$$

An extra $O(2)$ symmetry within the $(\mathbf{e}_1, \mathbf{e}_2)$ vector space is immediately apparent. The lowest non-trivial representation, U for $SO(3)$ is three-dimensional. Therefore, by introducing,

- A third orthonormal vector $\mathbf{e}_3 = \mathbf{e}_1 \times \mathbf{e}_2$.
- A projector $P \sim \text{diag}(1, 1, 0)$ to the $(\mathbf{e}_1, \mathbf{e}_2)$ plane.
- Defining $S_0 = (\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ and performing a gauge transformation $U = U S_0$ we can re-write,

$$\begin{aligned} H &= -\frac{3JS^2a^2}{2} \text{Tr} (S_0^{-1} P U^{-1} \nabla^2 U P S_0) \\ &= -\frac{3JS^2a^2}{2} \text{Tr} (P U^{-1} \nabla^2 U) \end{aligned} \quad (5)$$

However, for a proper RG treatment of the action it is necessary to consider a separate coupling constant [5]. This introduces a second coupling to the problem with the complete hamiltonian now given by,

$$\begin{aligned} H &= -p_1 (\mathbf{e}_1 U^{-1} \cdot \nabla^2 U \mathbf{e}_1 + \mathbf{e}_2 U^{-1} \cdot \nabla^2 U \mathbf{e}_2) - p_2 (\mathbf{e}_3 U^{-1} \cdot \nabla^2 U \mathbf{e}_3) \\ &= -\text{Tr} (P (U^{-1} \nabla U)^2) \end{aligned} \quad (6)$$

with gauge transformation $U = U(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ generating the full $SO(3)$ Goldstone modes and the diagonal coupling matrix $P = \text{diag}(p_1, p_1, p_2)$ ensuring the $O(3) \times O(2)/O(2)$ nature of the excitations. Given $\pi_1(SO(3)) = Z_2$, the order parameter U can have Z_2 point defects which are gapped out at zero temperature. Therefor the quantum action only has terms with second time derivatives and the full Goldstone mode action is therefore given by [4, 6].

$$S = - \int d\tau \int d^d x [\text{Tr} (P_\tau (U^{-1} \partial_\tau U)^2) + \text{Tr} (P_x (U^{-1} \nabla U)^2)] \quad (7)$$

where the two distinct diagonal matrices, P_τ and P_x presupposes the existence of three different spin-wave velocities of the problem as also obtained from a spin-wave expansion around the ordered state.

II. BOND-DISORDERED NON-COLLINEAR ANTIFERROMAGNET

First, we consider an isolated impurity within one of the elementary plaquettes and we find that the additional contribution from a bond impurity across sublattices AB , $J_{AB} = J + \delta J$ is given by,

$$\delta H = \delta J \mathbf{S}_A(\mathbf{x}_A) \cdot \mathbf{S}_B(\mathbf{x}_B) \quad (8)$$

. Unlike the bulk, the energy cost from Goldstone mode excitations now has contribution from the uncompensated single gradient term. Like before, we use the convenient orthonormal basis transformation and find the impurity contribution to be of the following form.

$$\begin{aligned} \delta H &= \delta J S^2 a \mathbf{n}_A U^{-1} \cdot (\mathbf{u}_{AB} \cdot \nabla) U \mathbf{n}_B \\ &= -\frac{\sqrt{3} \delta J S^2 a}{4} (\mathbf{e}_2 U^{-1} \cdot (\mathbf{u}_{AB} \cdot \nabla) U \mathbf{e}_1 - \mathbf{e}_1 U^{-1} \cdot (\mathbf{u}_{AB} \cdot \nabla) U \mathbf{e}_2) \end{aligned} \quad (9)$$

where we have exploited the orthonormality of the basis to drop the terms $\mathbf{e}_1 U^{-1} \nabla U \mathbf{e}_1$ and $\mathbf{e}_2 U^{-1} \nabla U \mathbf{e}_2$ which are zero. For a bond impurity across a bond orientation $\mathbf{u}_\alpha$ the perturbation assumes the form,

$$\begin{aligned} \delta H &= h^\alpha \mathbf{u}_\alpha \cdot \text{Tr} \left(P_x t^3 \begin{pmatrix} \mathbf{e}_1^T \\ \mathbf{e}_2^T \\ \mathbf{e}_3^T \end{pmatrix} U^{-1} \nabla U (\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3) \right) \\ &= h^\alpha \mathbf{u}_\alpha \cdot \text{Tr} (P_x t^3 U^{-1} \nabla U) \end{aligned} \quad (10)$$

. The same gauge choice and projector from the bulk action was used and the $SO(3)$ generator corresponding to the in plane rotation

$$t^3 = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (11)$$

makes an automatic appearance. This tells us that in the triangular lattice the local dipolar field arising due to bond disorder couples directly with the Goldstone mode associated with the in plane rotation. Now, it is straightforward to generalize to the case of spatially distributed bond impurities i.e. disordered bonds.

$$H = \int d^d x [\text{Tr} (P_x (U^{-1} \nabla U)^2) + h(\mathbf{x})^\alpha \mathbf{u}_\alpha \cdot \text{Tr} (P_x t^3 (U^{-1} \nabla U))] \quad (12)$$

III. REPLICA RENORMALIZATION GROUP

We consider a gaussian distribution for the disordered couplings,

$$\mathcal{P}[h] = \int \mathcal{D}h e^{-\frac{1}{2\Delta^2} \int d^d x h^\alpha(\mathbf{x}) h^\alpha(\mathbf{x})} \quad (13)$$

which allows us to use the replica trick and write down a homogeneous action suitable for the bond-disordered non-collinear order, albeit, at the expense of added replica indices.

$$\begin{aligned} S = & - \sum_r \int d\tau \int d^d x \text{Tr} \left[P_\mu (U_r^{-1} \partial_\mu U_r)_G^2 \right] \\ & - \Delta^2 \sum_{r,s} \int d\tau \int d\tau' \int d^d x \text{Tr} \left[P_x t^3 (U_r^{-1} \nabla U_r)_G \right] \cdot \text{Tr} \left[P_x t^3 (U_s^{-1} \nabla U_s)_G \right] \end{aligned} \quad (14)$$

Where the concrete example for the triangular lattice Heisenberg antiferromagnet shall have $\mathcal{L}[G] = \mathcal{L}[SO(3)]$ and $t^3 \in \mathcal{L}[SO(2)]$. We perform a trivial rescaling to restore the space-time Lorentz symmetry and re-write conveniently,

$$\begin{aligned} S = & - \sum_r \int d\tau \int d^d x \left(\frac{g_1}{2} \text{Tr} \left[(U_r^{-1} \partial_\mu U_r)_{G/X}^2 \right] + \frac{g_2}{2} \text{Tr} \left[(U_r^{-1} \partial_\mu U_r)_X^2 \right] \right) \\ & - \frac{\Delta^2 g_2^2}{4} \sum_{r,s} \int d\tau \int d\tau' \int d^d x \text{Tr} \left[t^3 (U_r^{-1} \nabla U_r)_G \right] \cdot \text{Tr} \left[t^3 (U_s^{-1} \nabla U_s)_G \right] \end{aligned} \quad (15)$$

with $X = SO(2)$. The dimensional natures of the bulk and disordered couplings are immediately apparent,

$$\begin{aligned} [g_{1,2}] &= d - 1 \\ [\Delta] &= (2 - d)/2 \end{aligned} \quad (16)$$

The bulk coupling $g_{1,2}$ is relevant and consistent with the fact that the homogeneous system is ordered. In this case $g_{1,2}$ has a trivial renormalization,

$$\begin{aligned} Z_{g_{1,2}} g_{1,2} \Lambda^{d-1} &= g_B \\ \implies \beta(g_{1,2}) &= (1 - d) \Lambda^{d-1} g_{1,2} + \mathcal{O}(1) \end{aligned} \quad (17)$$

In $d = 1$, the bulk coupling becomes marginal and disorder is relevant. Disorder shifts the wilsonian fixed point of the bulk action. However, in the following we shall focus on $d \geq 2$ and try to track the renormalization of the disorder coupling with the help of a perturbative expansion of the Goldstone mode action.

We use the exponential representation for the full symmetry group, $U = e^{\xi \cdot \mathbf{t}}$ with $t^a \in \mathcal{L}[G]$ and ξ_a are the associated Goldstone modes. We identified the in-plane Goldstone mode $\in X$ to be special and there is an easy separation between the respective generators of $\mathcal{L}(G/X)$ and $\mathcal{L}(X)$. Using the notation,

$$\begin{aligned} t^a &\in \mathcal{L}(G/X) \\ t^\alpha &\in \mathcal{L}(X) \end{aligned} \tag{18}$$

we have the commutation relations,

$$\begin{aligned} [t^\alpha, t^\beta] &= f_{\alpha\beta\gamma} t^\gamma \\ [t^\alpha, t^b] &= f_{\alpha bc} t^c \\ [t^a, t^b] &= f_{ab\gamma} t^\gamma \text{ and} \\ \text{Tr} [t^a t^b] &= -\delta^{ab} \end{aligned} \tag{19}$$

With the Lie algebra defined it is now straightforward to expand the terms of the action in terms of the Goldstone modes.

$$\begin{aligned} -\text{Tr} [(U^{-1} \partial_\mu U)_{G/X}^2] &= (\partial_\mu \xi_a)^2 + \frac{1}{3} (f_{ab\gamma} \xi_a \partial_\mu \xi_b)^2 + \dots \\ -\text{Tr} [(U^{-1} \partial_\mu U)_X^2] &= (\partial_\mu \xi_\alpha)^2 \\ -\text{Tr} [t^\alpha (U^{-1} \partial_\mu U)_G] &= \partial_\mu \xi_\alpha + \frac{K_{bc\alpha}}{2!} (\partial_\mu \xi_b) \xi_c + \frac{K_{bcd\alpha}}{3!} (\partial_\mu \xi_b) \xi_c \xi_d + \dots \end{aligned} \tag{20}$$

where we have used the fact that $X = SO(2)$ is an abelian group and used a compact notation $K_{bc\alpha} = [t^b, t^c]_\alpha = f_{bc\alpha}$, $K_{bcd\alpha} = [[t^b, t^c], t^d]_\alpha = f_{bc\alpha} f_{d\alpha}$ and so on. The explicit form of the action therefore is.

$$\begin{aligned} S &= \sum_r \int d\tau \int d^d x \left(\frac{g_1}{2} [(\partial_\mu \xi_a^r)^2 + \frac{1}{3} (f_{ab\gamma} \xi_a^r \partial_\mu \xi_b^r)^2 + \dots] + \frac{g_2}{2} [(\partial_\mu \xi_\alpha^r)^2] \right) \\ &\quad - \frac{\Delta^2 g_2^2}{4} \int d^d x \left[\sum_r \int d\tau \left(\partial_\mu \xi_\alpha^r + \frac{K_{bc\alpha}}{2!} (\partial_\mu \xi_b^r) \xi_c^r + \frac{K_{bcd\alpha}}{3!} (\partial_\mu \xi_b^r) \xi_c^r \xi_d^r + \dots \right) \right]^2 \end{aligned} \tag{21}$$

It is interesting to note the nature of infrared divergence of the theory, for the $\xi_\alpha^r \in SO(2)$ bosons, driven by the quenched disorder term of the action. One way to regulate this divergence is to introduce an external field term which should ultimately be taken to zero [7]. Introducing field strength, coupling constant and disorder strength renormalization constants, Z , $Z_{g_{1,2}}$ and Z_Δ we write down the full renormalizable replica

theory for our problem.

$$\begin{aligned}
S_{ren} = & \sum_r \int d\tau \int d^d x \left(Z_{g_1} \Lambda^{d-1} \frac{g_1}{2} \left[Z(\partial_\mu \xi_a^r)^2 + \frac{1}{3} (Z f_{ab\gamma} \xi_a^r \partial_\mu \xi_b^r)^2 + \dots \right] + Z_{g_2} \Lambda^{d-1} \frac{g_2}{2} [Z(\partial_\mu \xi_\alpha^r)^2] \right) \\
& - Z_\Delta^2 Z_{g_2}^2 \Lambda^d \frac{\Delta^2 g_2^2}{4} \int d^d x \left[\sum_r \int d\tau \left(\sqrt{Z} \partial_\mu \xi_\alpha^r + Z \frac{K_{bc\alpha}}{2!} (\partial_\mu \xi_b^r) \xi_c^r + Z^{3/2} \frac{K_{bcd\alpha}}{3!} (\partial_\mu \xi_b^r) \xi_c^r \xi_d^r + \dots \right) \right]^2 \\
& - Z^{-1/2} \Lambda^{d-1} \frac{g_2}{2} m^2 \sum_r \int d\tau \int d^d x \left[1 - \frac{Z}{2!} (\xi_\alpha^r)^2 + \frac{Z^2}{4!} \text{Tr} [t^\alpha t^\alpha t^\alpha t^\alpha] (\xi_\alpha^r)^4 \dots \right]
\end{aligned} \tag{22}$$

We consider the Greens function for the ξ_α^r Goldstone modes which gives rise to the logarithmic divergence of the theory.

$$G_\alpha^{rs}(\omega, \mathbf{k}) = \frac{1}{Z_{g_2} Z \Lambda^{d-1}} \frac{1}{g_2} \frac{\delta^{rs}}{\omega^2 + \mathbf{k}^2 + m_0^2} + \frac{Z_\Delta^2 \Lambda^{2-d}}{Z} \frac{\Delta^2}{2} \frac{\mathbf{k}^2}{(\omega^2 + \mathbf{k}^2 + m_0^2)^2} 2\pi \delta(\omega) \tag{23}$$

with,

$$m_0^2 = Z^{1/2} \Lambda^{d-1} \frac{g_2}{2} m^2 \tag{24}$$

The two-point function for ξ_α^r bosons up to one loop is,

$$\begin{aligned}
\Gamma_{rs}^{(2)}(\omega, \mathbf{k}) = & Z_{g_2} Z \Lambda^{d-1} g_2 (\omega^2 + \mathbf{k}^2) \delta_{rs} - Z_\Delta^2 Z Z_{g_2}^2 \Lambda^d \frac{\Delta^2 g_2^2}{2} \mathbf{k}^2 \\
& + Z^{1/2} \Lambda^{d-1} g_2 m^2 \delta_{rs} - Z^{3/2} \Lambda^{d-1} g_2 m^2 \delta_{rs} \frac{Z_\Delta^2 \Delta^2}{8} \Lambda^{2-d} \int \frac{d^d k}{(2\pi)^d} \frac{\mathbf{k}^2}{(\mathbf{k}^2 + m_0^2)^2}
\end{aligned} \tag{25}$$

To capture the leading logarithmic contribution we perform dimensiona regularization and use the following formulae,

$$\begin{aligned}
\int \frac{d^d k}{(2\pi)^d} \frac{1}{\mathbf{k}^2 + m_0^2} &= \frac{(m_0)^{d-2}}{(4\pi)^{d/2}} \Gamma(1 - d/2) \\
\left(\frac{4\pi \Lambda^2}{m_0^2} \right)^{\epsilon/2} &= 1 + \frac{\epsilon}{2} \log \left(\frac{4\pi \Lambda^2}{m_0^2} \right) + \mathcal{O}(\epsilon^2) \\
\Gamma(-n + \epsilon) &= \frac{(-1)^n}{n!} \left[\frac{1}{\epsilon} + \psi(n+1) \right] + \mathcal{O}(\epsilon)
\end{aligned} \tag{26}$$

with $\epsilon = 2 - d$. Now using the standard renormalization conditions to make the two point function finite

we obtain.

$$\begin{aligned}
Z &= 1 + \left(\frac{\Delta^2}{2}\right) \frac{1}{\epsilon} \\
Z_{g_2} &= 1 - \left(\frac{\Delta^2}{2}\right) \frac{1}{\epsilon} \\
Z_\Delta &= 1 + \left(\frac{\Delta^2}{4}\right) \frac{1}{\epsilon}
\end{aligned} \tag{27}$$

The renormalization of the disorder strength Δ follow straightforwardly by considering,

$$\begin{aligned}
Z_\Delta^2 \Lambda^\epsilon \Delta^2 &= \Delta_B^2 \\
\frac{d(Z_\Delta^2 \Delta^2)}{d \log \Lambda} &= -\epsilon Z_\Delta^2 \Delta^2 \\
\frac{d\Delta^2}{d \log \Lambda} &= -\frac{\epsilon}{\frac{d \log(Z_\Delta^2 \Delta^2)}{d \Delta^2}} \\
\frac{d\Delta^2}{d \log \Lambda} &= -\epsilon \Delta^2 + \frac{\Delta^4}{4} \\
\frac{d\Delta^2}{d \log \Lambda} &= (d-2)\Delta^2 + \frac{\Delta^4}{4}
\end{aligned} \tag{28}$$

The trivial renormalization of the bulk coupling has already been mentioned above and the subleading logarithmic correction is unimportant in this case. An interesting aspect of the calculation is that the renormalization was driven by the in plane Goldstone mode fluctuation and therefore the results shall be identical for the case of planar non-collinear antiferromagnets, i.e the XY case.

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- [1] Imry and Ma, 75
 - [2] Mermin and Wagner, 66
 - [3] Aharony, 78
 - [4] Azaria, Delamotte and Mouhanna, 92
 - [5] Azaria, Delamotte, Delduc and Jolicœur, 93
 - [6] Dombre and Read, 88
 - [7] Brezin and Zinn-Justin, 76